Lesson notes · for you

Data engineering: unit check

Data engineering · Level 4 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Schemas and formats
- ETL with Python and SQL
- Data quality and contracts
- Batch and stream

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (15 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (8 min): Reveal each answer and ask one team to explain why it's right.

Discussion

- Which part of a pipeline fails most often, and why?
- What would you put in a contract for data you depend on?
- Which system you use should be a stream, and isn't?

Check yourself

- One design decision I'd now make differently, and why.

Try it this week

Pick one dataset you know and design its whole path: the schema, the format, an idempotent load, three contract tests and batch or stream.

Worksheet

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Name _____ Class _____ Date _____

1 **In a star schema, where does each school's name belong?**

◯ Once, in the schools dimension table ◯ In every row of the fact table ◯ In the run log

2 **JSON is usually more than twice the size of CSV for the same table, because it repeats every column's name.**

◯ True ◯ False

3 **Put the steps of a safe nightly load in order.**

☐ Save the new watermark and log

☐ Delete the day's rows, then insert

☐ Transform them

☐ Extract the newer rows

☐ Read the watermark

4 **Fill the gaps: an idempotent load of today's rows.**

```
_________ FROM daily WHERE day = 'Mon';
_________ INTO daily SELECT * FROM today;
```

Word bank: `INSERT` `DELETE` `SELECT` `UPDATE`

5 **Match each quality test to what it catches.**

Range	• • • •	The same reading loaded twice
Referential integrity	• • • •	Today's file arriving at 09:30, not 07:00
Freshness	• • • •	A mark for pupil 7 when there's no pupil 7
Uniqueness	• • • •	A mark of 25 out of 20